Software Specification

for

CareLog G25

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version
WNG, MRS, NNWY, WKJ	31/08/2025	Initial Software Proposal	1.0 - Concept
WNG	10/09/2025	Interviews Added, Justification Format added	1.1 - Interview
MRS, NNWY, WNG	14/09/2025	System Context Analysis completed Use case diagram completed, Extra Sections for NFRs added	1.2 - SCA - Use case diagram
WNG, MRS, NNWY, WKJ	05/10/2025	Feedback Read and Revisions implemented, FR, NFRs sorted once more, removed out of scope AC's	2.0 - Revisions
NNWY	05/10/2026	Added Class Diagrams	2.1 - Class Diagrams
MRS	06/10/2025	Added Sequence Diagrams	2.2 - Sequence Diagrams
NNWY	06/10/2025	Added User Evaluation	2.3 - User Evaluations

1. Introduction

1.1 Purpose

This document specifies the software requirements for the Carelog software, a python based software solution designed for hospital staff and patients to log notes about patients. The purpose of the log is to help medical staff track a patient's overall experience and cultivate patient-centered empathy. The log allows the system to record both patient standard clinical observations (e.g., eating, medical diagnoses) and patient personal insights such as cultural background and emotional concerns.

1.2 Intended Audience and Reading Suggestions

This document is intended primarily for use by project managers, documentation writers, developers and testers/QA. It contains within target functionality that is to be developed by developers and their project managers, alongside acceptance criteria for testers. A different, simplified version of this document should be created and formatted as an End User Manual (EUM) and/or as a readme file.

The suggested reading (and order of reading) for each of these groups is :
Project Leads/Managers :

- Introduction - Problem Statement - Stakeholder Analysis - System Features/Requirements and their respective subsections - Use Case Diagram, System Context Analysis - System Architecture

Developers :

- Introduction - Problem Statement - System Features & Requirements - Use Case Diagram, System Context Analysis - System Architecture

Software Testers/Quality Assurance :

- Introduction - Problem Statement - System Features (Subsection FFR and Acceptance Criteria) - Use Case Diagram - System Context Analysis

Documentation Writers :

- Introduction - Problem Statement - Stakeholder Analysis - System Features - Use Case Diagram - System Context Analysis - System Architecture

2. Problem Statement

Hospitals and care centers have increasingly recognized that patient care extends beyond clinical observations but to include emotional and cultural factors as they play a major role in a patient's wellbeing.

However, current logging systems used by medical staff are only designed to record patient's clinical records such as their diagnosis report, medications and vital signs. These systems do not record the more personal aspects of patients such as their preferences and opinions on their medical care.

The lack of humanistic care can result in gaps in communication between staff and patients, reduced trust between patients and medical personnel. For patients from diverse backgrounds, the absence of a personalised logging tool to communicate with medical staff will allow them to feel neglected and concerned with their foreign medical care environment,

Therefore, there is a need for a lightweight software solution that aims to enable staff and patients to record not only standard clinical results but also relevant personal insights. This would promote empathetic care by ensuring that patient's voices and experiences are documented alongside clinical observations. CareLog (Culturally Aware, Reflective, Empathetic Logging) addresses this need by supporting ethical and secured design principles while maintaining usability for hospital staff.

3. Stakeholder Analysis

The CareLog system has several stakeholders and actors who either directly interact with the system or benefit from its outputs. Each stakeholder plays a different role ensuring that the system meets its objectives of a personalised and empathetic patient care software solution.

Primary Users:

Medical staff (Doctors and Nurses)
Role: To log clinical observations and record personal insights of patients such as cultural and emotional needs.

Interest:
Ensure patient data is accurate and easily accessible to improve treatment and care quality.

Patients
Role: To indicate preferences such as personal habits and cultural needs to medical staff to be recorded in the CareLog system.

Interest:
Feel their opinions are being heard and are further involved in their own medical treatment and care.

Indirect Users:

Administrative staff

Role: To audit patient logs for completeness and assign patients to specific care staff.

Interest:
Wants efficient information retrieval process and easy management of system-wide data access and storage.

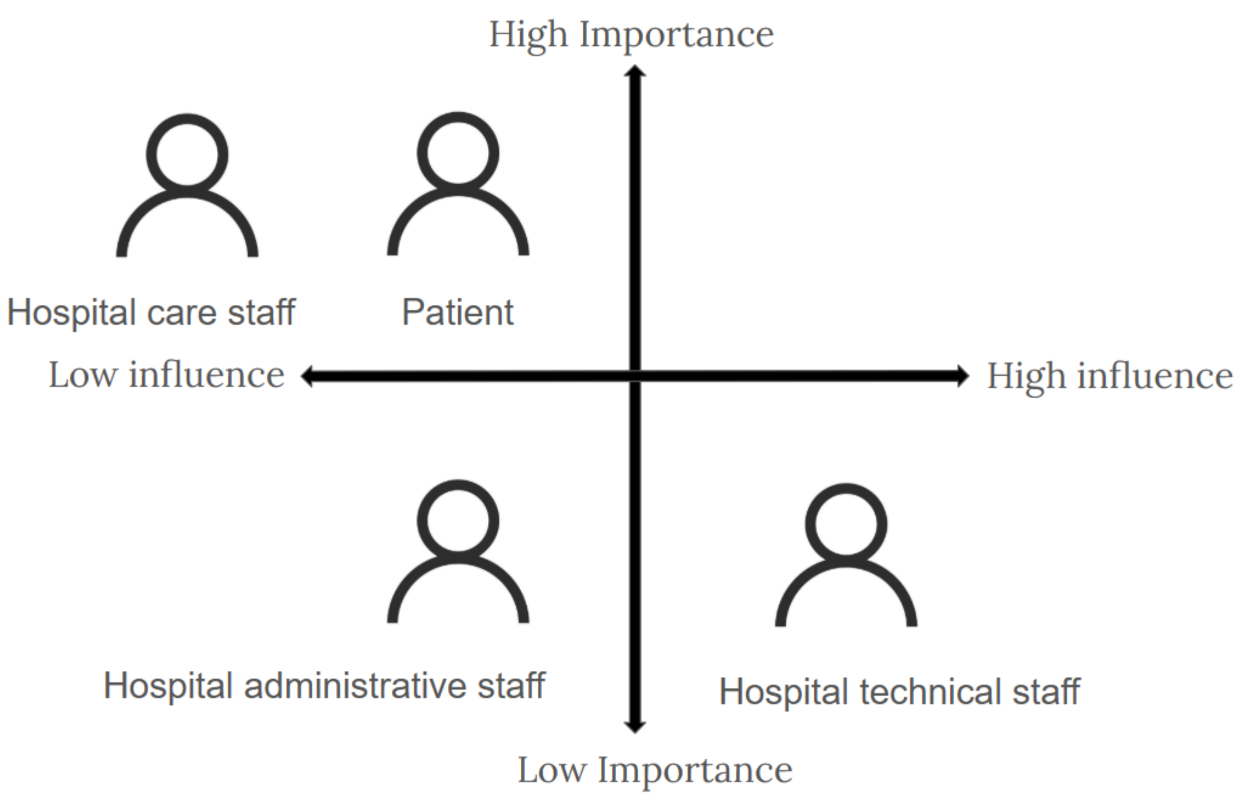
Technical Staff

Software engineer

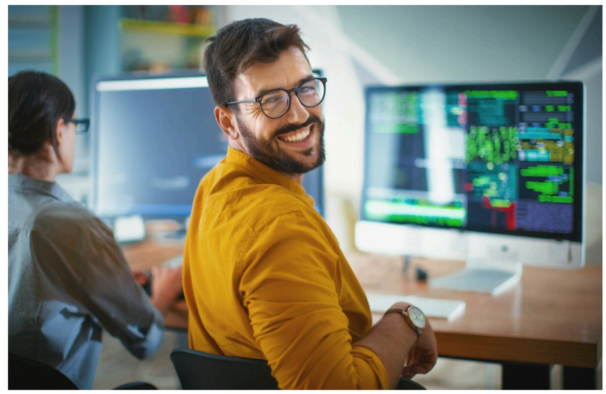
Role: Responsible for developing and maintaining the operation of the CareLog System, ensuring data security and system availability.

Interest:
To deliver a reliable, secure and scalable software system that meets the requirements required and comply with the relevant regulations.

Stakeholder Important / Influence Matrix:



Persona 1

Item	Description
Name	Joshua Doe
Picture	 [1]
Role	Software engineer that is responsible of developing the CareLog system
Demographic data	22, junior software engineer
Core characteristics	Dedicated, introverted
Core goals	To build functional and efficient software that meet user requirements and solve real world problems.
Typical challenges	To work effectively with the developing team under tight deadlines while needing to keep up in pace with the constantly evolving project requirements.
Singularities	Team player, highly skilled
Working situation	High stress working environment, able to work from home but has to be always on call, have constant meetings with project manager to ensure project direction is aligned with the user requirements, identifying and fixing bugs in the software
Place of work	Office, home
Expertise	Experienced in using Python, Java, Javascript, C++ and SQL
Main tasks with the system support	To develop functional and efficient CareLog software
Most Important tasks	Ensure usability, security and performance of the system.
Least important tasks	Producing a user manual for the CareLog software.

Persona 2

Item	Description
Name	Frank Brown
Picture	 [2]
Role	Hospital staff using the CareLog system
Demographic data	53, Consultant doctor
Core characteristics	Dedicated, compassionate
Core goals	To treat patients
Typical challenges	Work life balance, handling difficult patients
Singularities	Caring, understanding
Working situation	High working load, poor work life balance, many patients to look after simultaneously, routine check up on patients
Place of work	Hospital
Expertise	Extensive medical knowledge, ability to accurately diagnose and treat illnesses and injuries
Main tasks with the system support	To record patient's clinical observations
Most Important tasks	To record patient's clinical observations
Least important tasks	To record patient's personal preferences and cultural habits

Persona 3

Item	Description
Name	Danny Jane
Picture	 [3]
Role	Patient using the CareLog system
Demographic data	65, retired worker
Core characteristics	Friendly, extroverted
Core goals	To live a healthy life
Typical challenges	Effective communication with hospital staff about their medical treatment, cultural needs being omitted
Singularities	Always willing to take their medicine on time
Working situation	Retired with a free and easy lifestyle, receive treatment at hospital ward
Place of work	Hospital treatment ward, home
Expertise	Able to use electronic devices
Main tasks with the system support	To provide insights on their personal preferences and cultural needs
Most Important tasks	Effective communication with hospital staff
Least important tasks	To view clinical observations

Persona 4

Item	Description
Name	Sabrina Will
Picture	 [4]
Role	Administrative staff using the CareLog system
Demographic data	34, Administrative Executive
Core characteristics	Detail oriented, supportive
Core goals	To support staff and management, ensure smooth operations of the hospital
Typical challenges	Managing multiple tasks simultaneously, handling difficult patients and hospital staff
Singularities	Ability to multitask without losing track of essential tasks
Working situation	High working load, many parallel tasks to accomplish
Place of work	Hospital administration counter
Expertise	Proficient with Microsoft Office Tools such as Access, Excel and Word. Expert user of current existing data logging software.
Main tasks with the system support	To record patient's complete data upon admission and assigning patients to specific care staff
Most Important tasks	Auditing patient logs for completeness and compliance, assign patients to specific care staff
Least important tasks	Manage system-wide data access and storage

3.1 Interview Snippets (Abridged)

3.1.1 Joshua Doe

Interviewer : Thank you for your time today, Joshua. We understand that you are the engineer that is in charge of this project, and its subsequent upkeep for the client, correct?

Joshua : That is correct, yes.

Interviewer : Very good. So, tell us, what is it that you wish to see from this project?

Joshua : Obviously I would very much like for the system to be accessible remotely via the internet, without needing to be on site all the time for maintenance. It's also imperative that it saves regularly with backups of all the files that can be activated in case of data corruption or drive failures.

Interviewer : Obviously, yes. Other than that, would you think of anything else you'd like out of the project?

Joshua : I think an important thing would be scalability. If the system cannot adequately handle hundreds of patient logs and logs, then it would be a very bad system [Chuckling]!

Subsection : Stakeholder Goals

- Remote access to database
- System Ease of Maintenance
- Regular Backups of Database in case of data loss
- Scalability and Reliability Concerns

3.1.2 Frank Brown

Interviewer : Greetings, and many thanks for making time in your schedule today, Dr. Brown. We understand that you are a very busy man, so we will keep this as short and brief as possible. We're asking a few key members of staff a few questions about what they'd like to see in the new CareLog system we are developing for the hospital. So to start with, I see here that you are carrying around a substantial amount of patient files, correct?

Dr. Brown : Uh, yes. That is correct.

Interviewer : Interesting. I won't pry, but I must ask, I presume you wish for this system to lessen the amount of papers and files needed during your consultation rounds, yes?

Dr Brown : That is correct, yes. I would also love to be able to tell at a glance what potential allergens a patient has, their cultural background and medical conditions on this application. It would massively reduce the amount of papers I have to flip[sic] on a daily basis to understand the patient's condition and what options are available at my disposal to treat their issues, without offending their cultural values or checking with them on their allergens.

Interviewer : Interesting. That is good to know, indeed….[ ]... I think that will be all we need from you, good doctor. That will be all for this interview, and we will obstruct your work no longer. Have a good day.

Subsection : Stakeholder Goals

- Ease of Access of Database
- Medical Records of Patients Accessible from CareLog (Including allergens etc.)
- Cultural Sensibilities of Patient Recorded (Knowing what treatments or care must be provided)

3.1.3 Danny Jane

Interviewer : Good afternoon, Mister..?

Danny Jane : Danny. Danny Jane. Mr Jane if you prefer, but I prefer Dan.

Interviewer : Dan it is, then. We're collecting opinions of randomly selected patients, and we'd like to hear your thoughts regarding some new software the hospital is developing. It's called "Carelog". It's an app that allows you to write personal logs and enter personal information for your Doctors to access when treating you.

Dan : I see, I see. So what do you need from me, then?

Interviewer : We would love to hear your thoughts about what you think we can add into this app. No judgements, we'd just love some suggestions from patients to develop the app to be more patient friendly.

Dan : Hmm. Interesting… [ ]... I suppose I would like for the app to be easy for me to use. I am 65 years of age, after all. Learning how to use a newfangled[sic] gadget like this.. It would be much easier if there were big, strong [icons] and big [fonts]. My eyesight is not what it used to be, you know.

Interviewer : Good.. Good… What about things such as your cultural background? Or perhaps dietary restrictions? Or… perhaps what medication you are taking. I spotted you getting some pills from the dispensary, no?

Dan : Ah! Yes! That is a very good point. I would love to be able to easily list out those things. But I would also like for something like.. A [text to speech function] in the app. Not all us old folk are as literate as you young ones!

Interviewer : Thank you, thank you. Your thoughts and feedback have been quite valuable. I've taken up enough of your time now, so I shall bother you no longer. Have a very nice day, Mr Jane.

Subsection : Stakeholder Goals

- Ease of Use of Application/Program (Accessibility, Minimal Training)
- Text to Speech Program for Input

3.1.4 Sabrina Will

(Editor Note : This was the persona we asked someone to assume. We did not record the interview, only a transcript.)

Sabrina : I understand this interview is about the CareLog software and what I wish to come from it, yes?

Interviewer : That would be correct, madame. I understand that your workload is quite immense, so I shall keep this as brief as possible. We would like to hear what you would like to see from the CareLog product.

Sabrina : First of all, I would absolutely need the software to be easy and intuitive to use. If it takes hours of training beforehand to get used to this new system, then it would take a large amount of our budget to do so. So, simple to use would be on the top of my priority list. Another top priority would be allowing the patient logs to be easily accessed by the hospital staff, allowing for us to gain a better understanding of their needs and conditions. In addition, it would allow us to assign specific care staff to the patient in case their culture or needs require as such.

Interviewer : Good to know.

Sabrina : I'm not done just yet, I also need the uploaded logs to be accessible to only appropriate staff, such as their primary care physicians or administrative staff with the appropriate credentials. The logs created by medical staff should not be allowed to be tampered with after their creation, to allow us to better track down medical fraud in the event such things do occur.

Interviewer : So to summarize, you need the software to be easy and intuitive to use, preferably a 'pick up and use' sort of software, Easy access to patient logs by appropriate hospital staff for easier assignment of primary caretaker or physicians, and most importantly an 'append only' system of medical records. Is that correct?

Sabrina : Precisely.

Interviewer : Alright, that should be all we need from you. Thank you very much.

Subsection : Stakeholder Goals

- Short training cycle for usage.
- Fast and quick access of logs, medical records and cultural needs for all appropriate medical staff.
- Restricted Access for logs to only appropriate staff.
- Append only editing for Medical Records
- Authentication of appropriate access levels to create or append medical records

4. System Features and Requirements and Acceptance Criteria

4.1 Features, Function Requirements and Non Functional Requirements

Functional Requirements

Log Creation, Uploading, Draft Saving Functionalities :

1. The software shall enable users to create logs which include log timestamp, log content and log author. [In Scope]

2. The system shall prompt the user with a confirmation warning message before sharing or uploading logs.[In Scope]

3. The system shall display a confirmation message to the user when the log is successfully saved to the database.[In Scope]

4. The system shall maintain append only behaviour for uploaded logs, meaning that uploaded logs cannot be edited or deleted by users. [In Scope]

5. The system shall maintain a database storage to store uploaded logs. [In Scope]

6. The system shall prevent orphan references by ensuring all uploading logs are linked to a valid user account. [In Scope]

7. The system shall maintain a local draft storage to store incomplete logs. [Out of Scope]

Login Functionality :

8. The system shall implement a Logon-Logoff mechanism to manage user sessions. [In Scope]

9. The system should differentiate between Normal Users (patients) and Medical Staff, restricting access to certain functionalities accordingly. [In Scope]

10. The system shall allow only authorized users to access or perform operations according to their assigned roles [In Scope]

11. The system shall alert the user when invalid login credentials are entered. [In Scope]

12. The application shall provide two factor authentication (2FA) for enhanced account security [Out of Scope]

13. The system should have “Forgot Password/Username” functionality that allows users to recover their account. [Out of Scope]

(Medical and Admin) Staff Record Functionality :

14. The system shall allow authorized medical staff to enter clinical observations on a patient, such as allergens, dietary restrictions or clinical test results. [In Scope]

15. The system shall only allow staff users with the respective staff access privileges to access staff only data (e.g registered patient accounts, staff accounts and patient clinical records) . [In Scope]

16. The system shall allow admin users to assign medical staff to care for patients. [In Scope]

17. The system shall allow admins with elevated privileges to perform registration and removal of user accounts for audit and compliance purposes. [In Scope]

18. The system shall ensure server-side integrity of patient records by preventing logs appended to invalid patients. [In Scope]

19. The system shall allow medical staff with respective privileges to view patient’s personal preferences and clinical observation [In Scope]

Patient Record Functionality:

20. The system shall allow the respective patient to view all own personal preference logs [In Scope]

21. The system shall allow the user to log personal preferences that will be saved to the database. [In Scope]

22. The system shall allow medical staff to save logs of patient’s personal preferences to the database [In Scope].

23. Patient Logs shall be timestamped and limited to a single log per day. [In Scope]

24. Patient Logs shall be append only once saved to the database [In Scope].

Non Functional Requirements

Performance

1. The system shall respond to user interactions (e.g.opening dashboards, loading log lists) within 3 seconds under normal load (<= 100 concurrent users) . [In Scope]
2. Uploading or saving load shall complete within 3 seconds on a stable connection
3. Staff only operations such as searching or appending patient logs shall return result within 3 seconds for datasets up to 10000 records
4. System performance shall remain stable with up to 100 concurrent sessions without data loss or timeout.

Reliability

5. The system should be able to maintain two separate servers holding the same data to maintain uptime during maintenance or major server problems. [Out of Scope]
6. The system should periodically back up its database to a different location to prevent data loss. [Out of Scope]
7. When saving a patient's log, if the log is incomplete or does not reference a specific patient to be associated with the log, flag it for deletion to prevent bad data or orphaned logs from entering the system. [Out of Scope]

Availability

8. The secondary server should be able to automatically take over if the primary server does not respond after 3 minutes of downtime. [Out of Scope]
9. The secondary server should upload all logs created during the primary server downtime back to the primary server once it is available once more. [Out of Scope]

Usability

10. The system should be easy and intuitive to use, such that an average user can operate core features with an hour or less of training. [In Scope]
11. The interface shall provide clear status messages, confirmations and warnings for every action performed. [In Scope]

Security

12. Data stored in the database shall be encrypted at rest using AES-256.
13. User sessions shall expire after 15 minutes of inactivity, requiring re-authentication. (Tentatively In Scope)
14. The system shall limit 5 login attempts before deactivating the account to mitigate brute force attacks. (Tentatively In Scope)
15. All actions shall require role validation on the client side.(Tentatively In Scope)
16. The system shall maintain an append only audit log of security relevant actions such as timestamp, author and content of recorded logs. (In Scope)
17. The system shall enforce single factor authentication which includes login with valid login credentials. (Tentatively In Scope)
18. The system shall optionally support Multi-Factor Authentication (MFA) via TOTP (e.g., Google Authenticator) for administrative and medical staff logins. (Out of Scope)

5 Acceptance Criteria

Login Functionality :

1. Given the user is opening the program, the user shall be prompted to login before accessing restricted functionalities.
2. When the user enters incorrect log in credentials,then the system shall display an error message:” Invalid username or password” and deny access (HTTP 401 Unauthorized) .
3. When the user attempts to access a restricted function(e.g remove patient) without the respective privileges, the system shall return an appropriate error (HTTP 403 Forbidden).
4. When the user enters valid login credentials, then the system shall login them in and load the interactive dashboard within 3 seconds. (NFR-1)
5. The system should be able to handle non-english alphabet usernames and handle UTF-8 encoding for all user credentials.
6. If the user remains idle for more than 15 minutes, then the system shall automatically log them out. (NFR-12)
7. When there are 5 continuous wrong login attempts, the user account will be deactivated. (NFR-13)

Log Creation, Uploading, Draft Saving Functionalities :

8. Given that a patient user is logged in, patients shall be able to create logs and view their own personal preference logs.
9. Given that a medical staff user is logged in, medical staff shall be able to create patient clinical observation and personal preference logs, view patient logs and assign care staff to patients. (FR14, FR16, FR19, FR22)
10. When the user uploads a log, then the system shall prompt for confirmation with a warning message stating that once uploaded, the log becomes immutable. (FR-2, FR-4
11. When the log is successfully uploaded or saved, the system shall return a visible confirmation message to the user interface. (FR-3)
12. The system shall link each uploaded log to a valid user account; uploads attached to an invalid user will be rejected with an error message. (FR-18)
13. The system shall complete all upload actions within 3 seconds. (NFR-2)
14. The system should save the log uploaded/created by the user to the correct account, and the log should only be accessible by that user unless it is shared. (FR-6)
15. The system shall not allow deletion or modification of logs by any user. (FR-4)

(Medical and Admin) Staff Record Functionality :

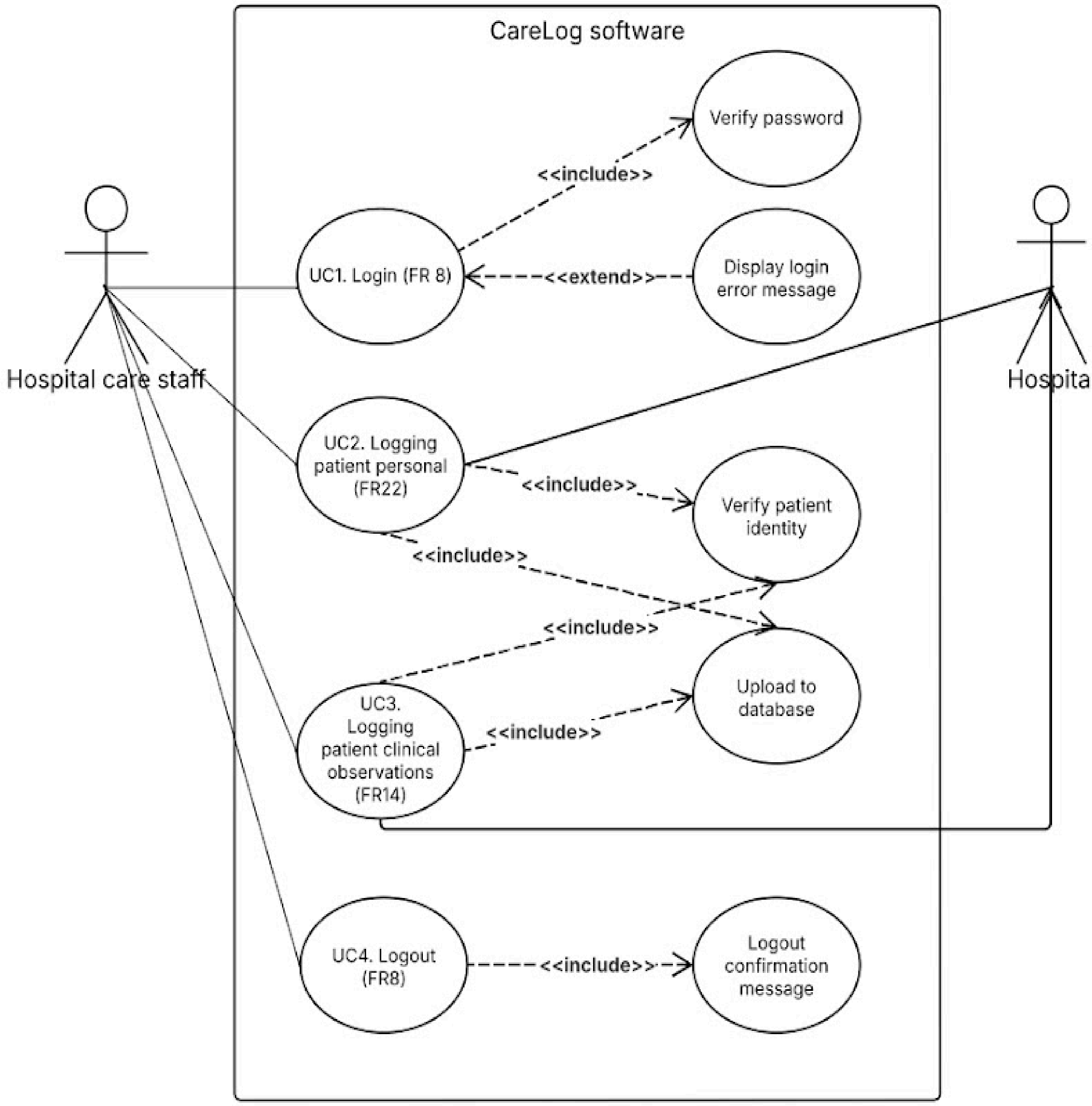
16. Given the user is logged in as medical staff, then they shall be able to access and append records for patients. (FR-9, FR-14, FR-22)
17. When a medical staff user appends a log to a patient record, then the system shall ensure a valid patient ID is associated. (FR6, FR-18)
18. If no content is entered in a new record, then the system shall reject the entry and display an error message.
19. When a new record is appended to the system, the system shall timestamp the entry and record who has created the new record. (FR-1)
20. When a medical staff user views patient logs, the user interface shall show all records associated with the patient. (FR-19)
21. If an unauthorized patient user attempts to view staff only data, then the system shall deny access and return 403 Forbidden.
22. All staff operations shall execute within <= 3 seconds for datasets of up to 10000 records. (NFR-3)
23. The user interface should clearly mark these as different from patient personal notes.(FR-9)
24. This information shall only be appendable and not editable to ensure data integrity. (FR-4)

Non Functional Requirements :

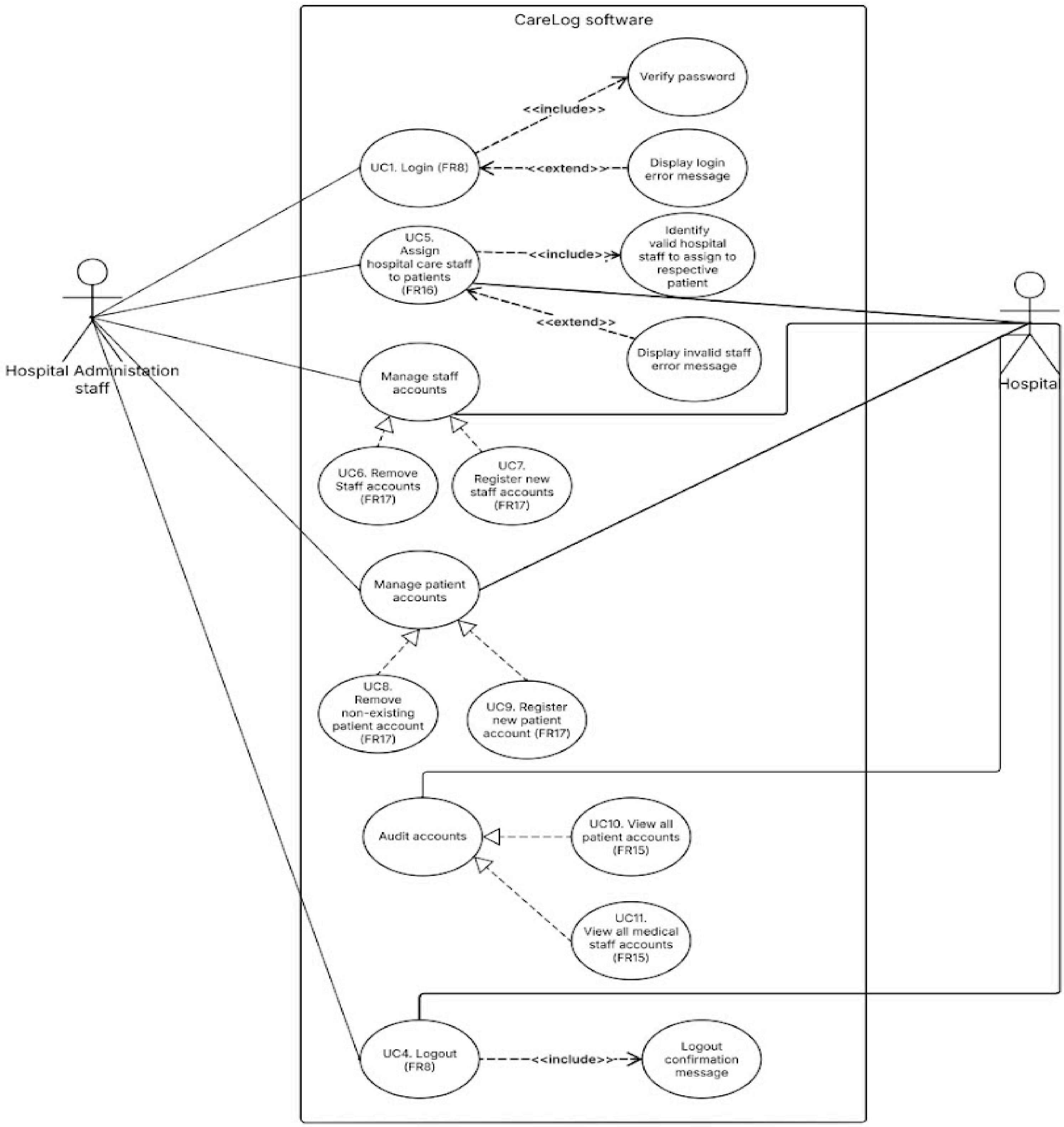
25. When using the interface, then users shall be able to navigate key functions (Login, Log Creation, Upload) with clearly labeled buttons and icons, requiring ≤ 1 hour of training. (NFR-8)
26. When using the interface, the user shall be aware of all actions that have been made. (NFR-9)

6 Use Case Diagram

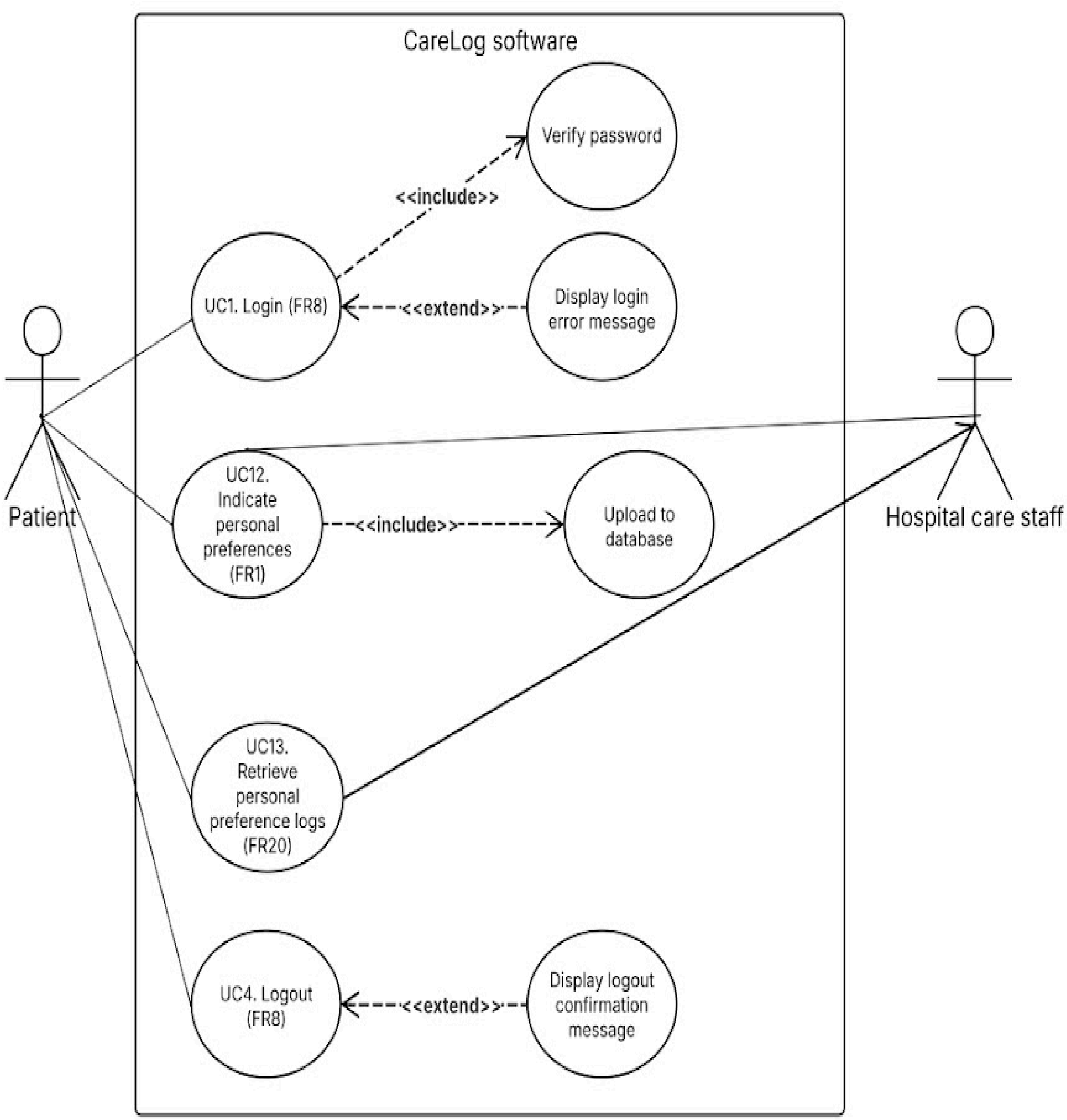
<u>Use case diagram for hospital care staff:</u>



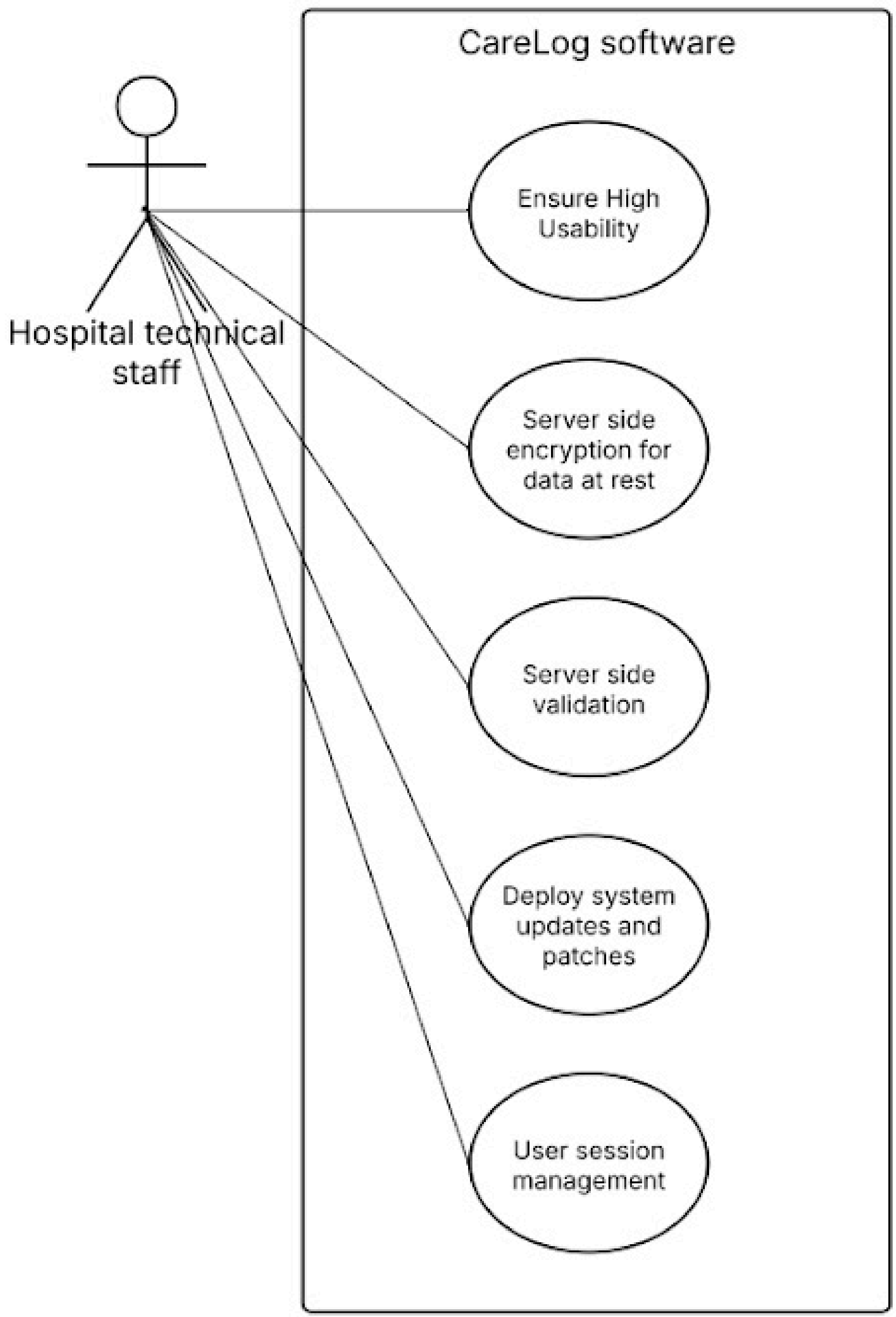
Use case diagram for hospital administration staff:



Use case diagram for hospital patients:



Use case diagram for hospital technical staff:



7 System Context Analysis

(All entries or otherwise marked with a * are out of scope for this build. See Assumptions and Out of Scope for further details.)

7.1 Overview

CareLog operates within a hospital environment to record both clinical observations and human-centred insights about patients. It must interoperate with hospital IT, respect privacy/ethics, and remain usable for staff and patients.

7.2 Operational Context

Primary users & locations. Doctors/nurses on wards, admin staff at desks, and patients at bedside or home (for post-discharge notes). Devices include ward desktops, nurses' tablets, and kiosk/bedside terminals.
Deployment. Web-first Python backend with a hospital hosted database, supports local draft saving on device with sync when online.

External systems :

***Hospital Identity Provider (IdP):** SSO/RBAC, differentiates staff and patient access.

***Secure Storage/Backup:** Automated off-site/secondary node backups for recovery.

***Notification service:** Email/SMS/in-app notices (e.g., when staff share a log).
Key operational qualities. Availability during rounds, responsive, append-only staff records, immutable uploaded logs, audit trails

EHR/HIS: Read patient demographics/identifiers, write CareLog references (not direct overwrites of clinical records).

Key operational qualities. Availability during rounds, responsive, append-only staff records, immutable uploaded logs, audit trails.

7.3 Behavioral Interaction Context

Typical flows.

1. Staff round → authenticate → open patient → view allergens/cultural notes → append staff record (append-only) → save.
2. Patient reflection → authenticate (or assisted mode) → write note → save as draft or upload → confirm sharing scope → immutable upload.
3. Admin tasks → assign staff to patients → review completeness/compliance dashboards → manage access issues.

   System reactions. Inline validation, role-gated visibility (staff-only fields), confirmations on upload/sharing, error messages during downtime with graceful sync.

7.4 Ecological Context

- ***Policies & law.** Complies with hospital policy and applicable privacy/data-protection laws (e.g., PDPA-style principles), including consent, purpose limitation, and data minimization.
- ***Interoperability.** Designed to integrate via common healthcare data standards where feasible (e.g., HL7/FHIR mappings for patient identifiers/links).
- **Security posture.** Encrypted transport, encrypted at rest for server data, strong authentication, routine backups and disaster-recovery procedures, production monitoring.
- **Operational constraints.** Network segmentation in hospitals, intermittent Wi-Fi on wards → local drafts and background sync.

7.5 Sociocultural Context

- **Human-centred intent.** Captures patient stories, preferences, cultural needs to support empathetic care and reduce miscommunication.
- **Accessibility.** Large fonts, clear language, option for text-to-speech/voice input, inclusive, multilingual UI where practical.
- **Cultural safety.** Sensitive fields (e.g., taboos, preferred caregiver gender) are visible on a need-to-know basis, explicit sharing confirmations, staff training notes can be linked in help.

- **Trust & agency.** Patients understand what is shared and with whom, logs uploaded by patients are immutable to preserve integrity, staff annotations are append-only with author and timestamp.

7.6 External Entities & Interfaces

Actors: Patients, Medical Staff, Admin Staff.
External Systems: *IdP/SSO, *EHR/HIS, *Notification Service, *Backup/DR *Store, Compliance/Audit Tools.
CareLog (System Boundary): *Authentication & RBAC, Log Editor (draft/upload), Staff Append Module, Sharing/Consent, Sync Service, *Audit/Reporting API.
Key Data Flows:

Patients ↔ CareLog: drafts, uploads, consent settings.

Staff ↔ CareLog: view patient profile, append staff notes, view patient logs.

***CareLog ↔ IdP:** authN/authZ tokens roles.

***CareLog ↔ EHR/HIS:** patient identifiers, read demographics, write back reference.

***CareLog ↔ Notification:** event messages.

***CareLog ↔ Backup/DR:** scheduled encrypted backups, restore on incident.

***CareLog ↔ Compliance:** audit exports/dashboards.

7.7 Assumptions & Out-of-Scope

Advanced security features (e.g., 2FA, active-active failover) and full cryptography implementations are design targets but out of scope for the first build, policies and procedures are documented for later phases.

Reliability features using a server are design targets, desirable for the carelog project to allow easy access for most hospital staff at any given moment, however for the first build, due to lack of infrastructure and technical knowledge, remain out of scope for the first build, and are included here as project documentation for potential future builds.

Direct editing of clinical records in the EHR is out of scope, CareLog links and augments with references only.

8 System Architecture

(All entries or otherwise marked with a * are out of scope for this build. See Out of Scope subsection for further details.)

8.1 Architectural Style

CareLog adopts a Layered Architecture model, which structures the system into multiple layers, each with distinct responsibilities.

Layers:

1. Presentation Layer
   - User interfaces (mobile app*, web portal).
   - Handles user interaction, input validation, accessibility features (e.g., large fonts, text-to-speech*)
2. Application Layer
   - Manages workflows (log creation, synchronization, notifications).
   - Coordinates between presentation and business logic.
3. Business Logic Layer
   - Encapsulates healthcare-specific rules and constraints.
   - Enforces role-based policies (patients, clinicians, admin).
   - Validates logs (append-only for staff, immutable uploads for patients).
4. Persistence Layer
   - Manages data storage and retrieval.
   - Supports cloud storage (NoSQL).*
   - Ensures append-only staff logs and immutable patient logs.
5. Infrastructure Layer
   - Provides supporting services such as authentication, encryption, backup*, integration with EHR/notification systems*.
   - Ensures compliance with hospital IT, privacy, and recovery policies.*

Rationale:

- Separation of Concerns: Each layer has a single responsibility, simplifying maintenance and testing.
- Security Enforcement: Role-based access and encryption applied consistently at the Business and Infrastructure layers.
- Scalability: Layers can scale independently (e.g., scaling application servers without affecting persistence).
- Maintainability: Easier to update or replace one layer (e.g., UI redesign) without disrupting others.
- Reliability: Infrastructure layer ensures backup*, redundancy*, and integration with external hospital systems*

8.2 Component and Module View

This section describes CareLog's main software building blocks (modules), their responsibilities, and how they interact to meet requirements.

Presentation Layer

- Login & Authentication UI
  - Supports login/logout.
  - Error messages for wrong credentials.
- Log Management UI
  - Create logs.
  - Upload logs with confirmation.
- Staff Dashboard
  - Append medical notes (allergens, restrictions, history).
  - Append-only records clearly marked separate from patient logs.
- Admin Dashboard
  - Assign staff, audit patient logs, manage user data.

Application Layer

- User Session Manager
  - Handles login state and role distinction (patient, staff, admin).
- Staff Record Manager
  - Enforces append-only entries for staff.
  - Validates required fields (no empty submissions).
- Audit & Access Manager
  - Controls which logs are visible to whom.
  - Applies discretionary sharing rules.

Business Logic Layer

- Rules Engine
  - Patients can upload logs but not edit after upload.
  - Staff logs are append-only, hidden from patients unless shared.
- Consent & Security Manager
  - Enforces patient-controlled sharing.
  - Applies AES encryption for uploads.

Persistence Layer

- Local Storage
  - Holds drafts and offline data.
- Central Database*
  - Stores uploaded logs, staff records, patient history.*
- Backup & Recovery System*
  - Periodic backups (scheduled off-peak).*

- Data recovery when required.*

Infrastructure Layer

Encryption Service

- Protects data in transit (uploads) and at rest (DB).

System Maintenance Module

- Ensures uptime, error messages during downtime, minimal disruption.

8.3 Deployment View

The Deployment View shows how CareLog's software components are distributed across devices and servers in the real-world environment.

Client Devices

- Mobile Devices (iOS/Android): Run the CareLog mobile app with local storage (drafts, offline logs).*
- Web Browsers (PC/Tablets): Access the CareLog web portal (admin + staff functions).

Application Server

- Hosts the main CareLog application (business logic, workflows).
- Handles user sessions, log uploads, staff record management, and access control.

Database Server

- Stores patient logs, staff append-only records, and account data.
- Centralized and secure, supporting both local and cloud access*.

Backup & Recovery Server*

- Stores encrypted backups of the database.*
- Ensures data recovery in case of system failure (scheduled during low-traffic hours).*

Authentication Service (Hospital IdP or CareLog Auth)

- o Validates user credentials (patients, staff, admins).
- o Supports role distinction for access control.

Notification Service

- o Sends reminders and alerts to patients and staff (via email, SMS, or push notifications).

8.4 Data Model and Storage

This section describes how data is organized, stored, and managed in CareLog, both locally (on devices) and centrally (in the database).

- User
  - Attributes: UserID, Username, Password (hashed), Role (Patient, Staff, Admin).
- Patient Log
  - Attributes: LogID, UserID, Content, Timestamp, Status (Draft/Uploaded).
  - Rules: Uploaded logs are immutable, drafts editable until uploaded.
- Staff Record
  - Attributes: RecordID, PatientID, StaffID, Content, Timestamp.
  - Rules: Append-only, cannot be edited once saved.
- Consent & Sharing
  - Attributes: LogID, SharedWith (UserID), AccessLevel.
  - Rules: Patient decides who can view shared logs.
- Audit Trail
  - Attributes: ActionID, UserID, ActionType, Timestamp.
  - Tracks key activities (log creation, uploads, staff entries).

Storage Layers:

- Local Storage (Mobile Devices)
  - SQLite/Realm database.
  - Holds drafts and offline logs until synced.
- Central Database (Server)
  - SQL/NoSQL database.
  - Stores all finalized logs, staff records, patient histories, and user accounts.
- Backup & Recovery System
  - Periodically stores encrypted copies of the central database.
  - Provides restoration capability in case of failure.

8.5 Security Architecture

The CareLog system prioritizes privacy, confidentiality, and controlled access to sensitive patient information.

Authentication & Authorization

- Secure login process with username/password.
- Role-based access control:
    - Patients: manage personal logs.
    - Staff: append medical notes (append-only).
    - Admin: elevated privileges (e.g., deletion).
- Unauthorized access attempts logged for auditing.

Encryption

- In Transit: All communication between client and server is encrypted (e.g., TLS/HTTPS).
- At Rest: Logs and records stored in the database are encrypted using AES.
- Drafts stored locally are also recommended to use device-level encryption.

Access Control

- Patient logs are private by default and only shared with staff when explicitly uploaded/shared.
- Staff notes remain hidden from patients unless proactively shared.
- Database queries strictly filtered based on user role and privileges.

Audit Trails

- Every action (log upload, staff append, admin deletion) is recorded in a tamper-resistant audit log.
- Audit logs are not editable or deletable.

Data Retention & Deletion

- Logs persist until explicitly deleted by the patient or an admin.
- Staff medical records are append-only and never deleted.

8.6 Out-of-Scope System Architecture Documentation

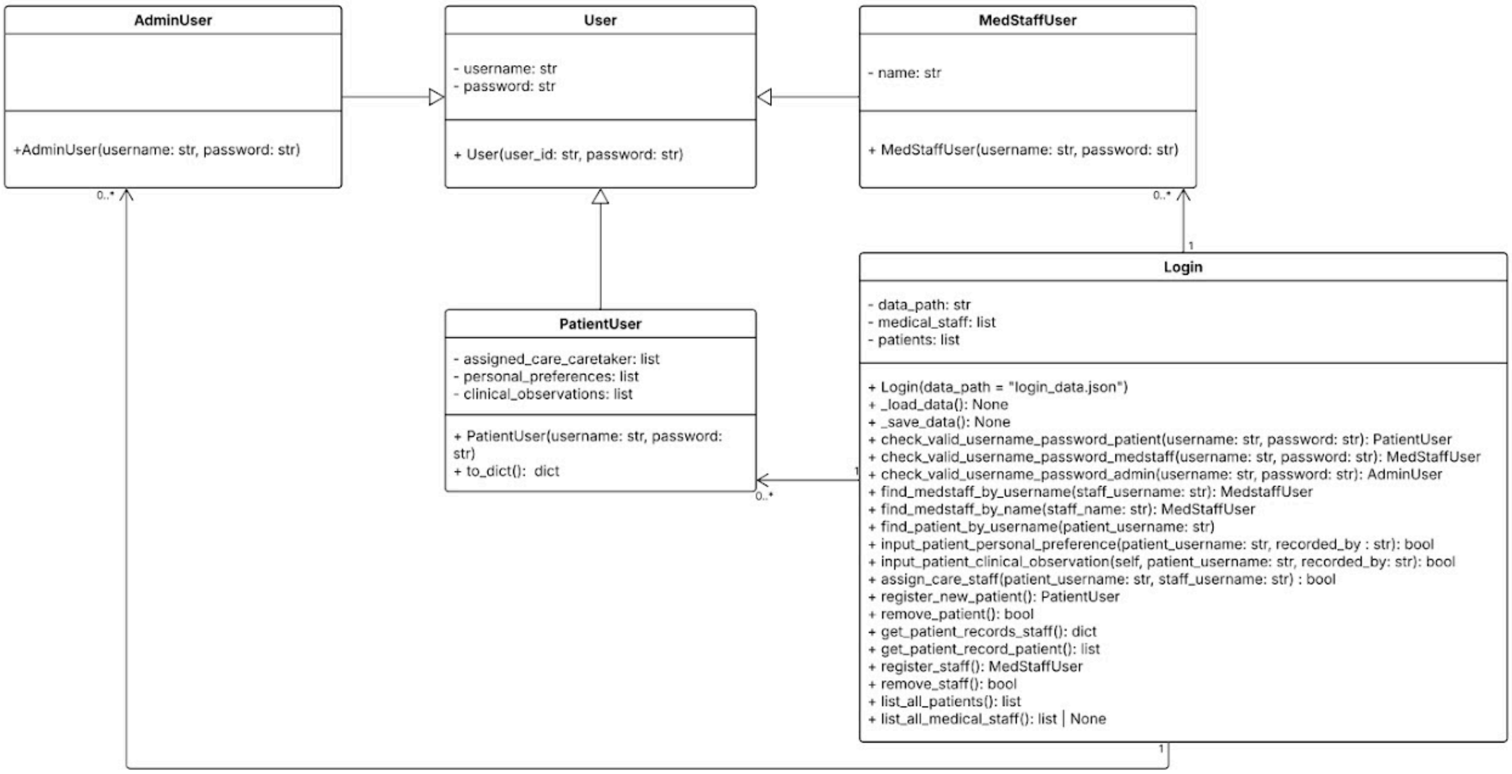
Advanced security features (e.g., 2FA, active-active failover) and full cryptography implementations are design targets but out of scope for the first build, only being documented for later potential phases.

Reliability features using a server are design targets, desirable for the carelog project to allow easy access for most hospital staff at any given moment, however for the first build, due to lack of infrastructure and technical knowledge, remain out of scope for the first build, and are included here as project documentation for potential future builds.

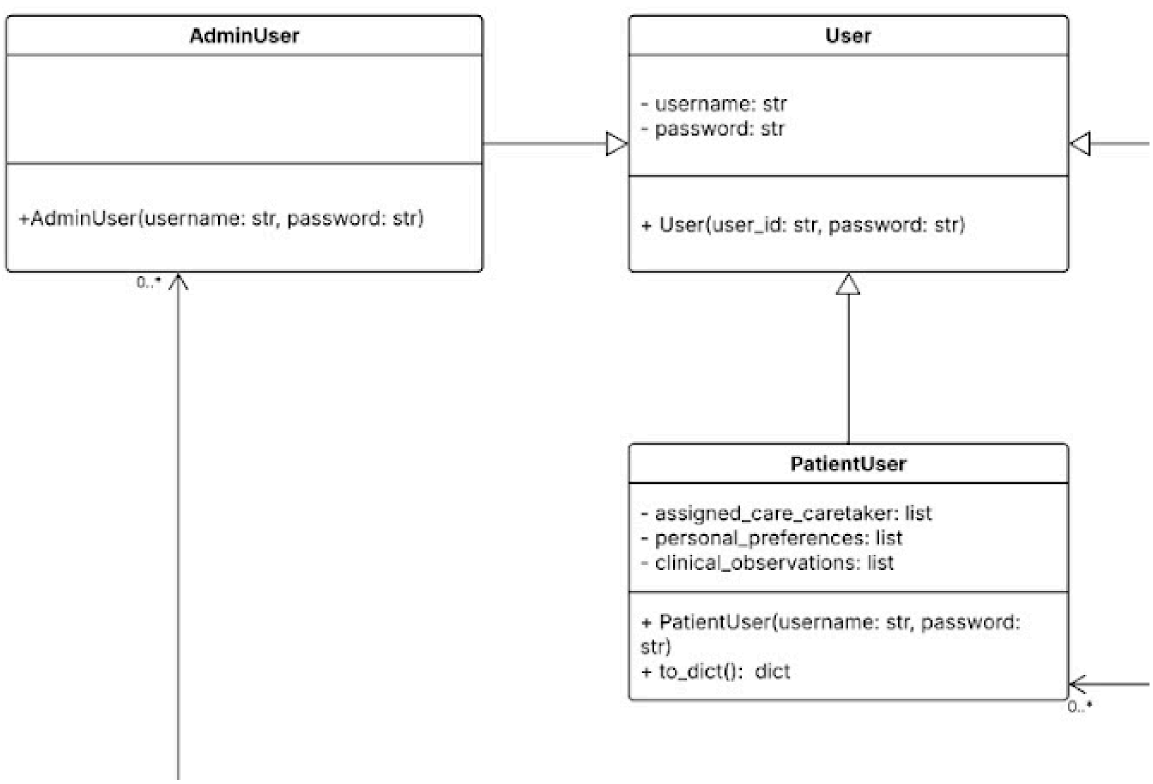
Direct editing of clinical records in the EHR is out of scope, CareLog links and augments with references only.

9. Class Diagram(s)

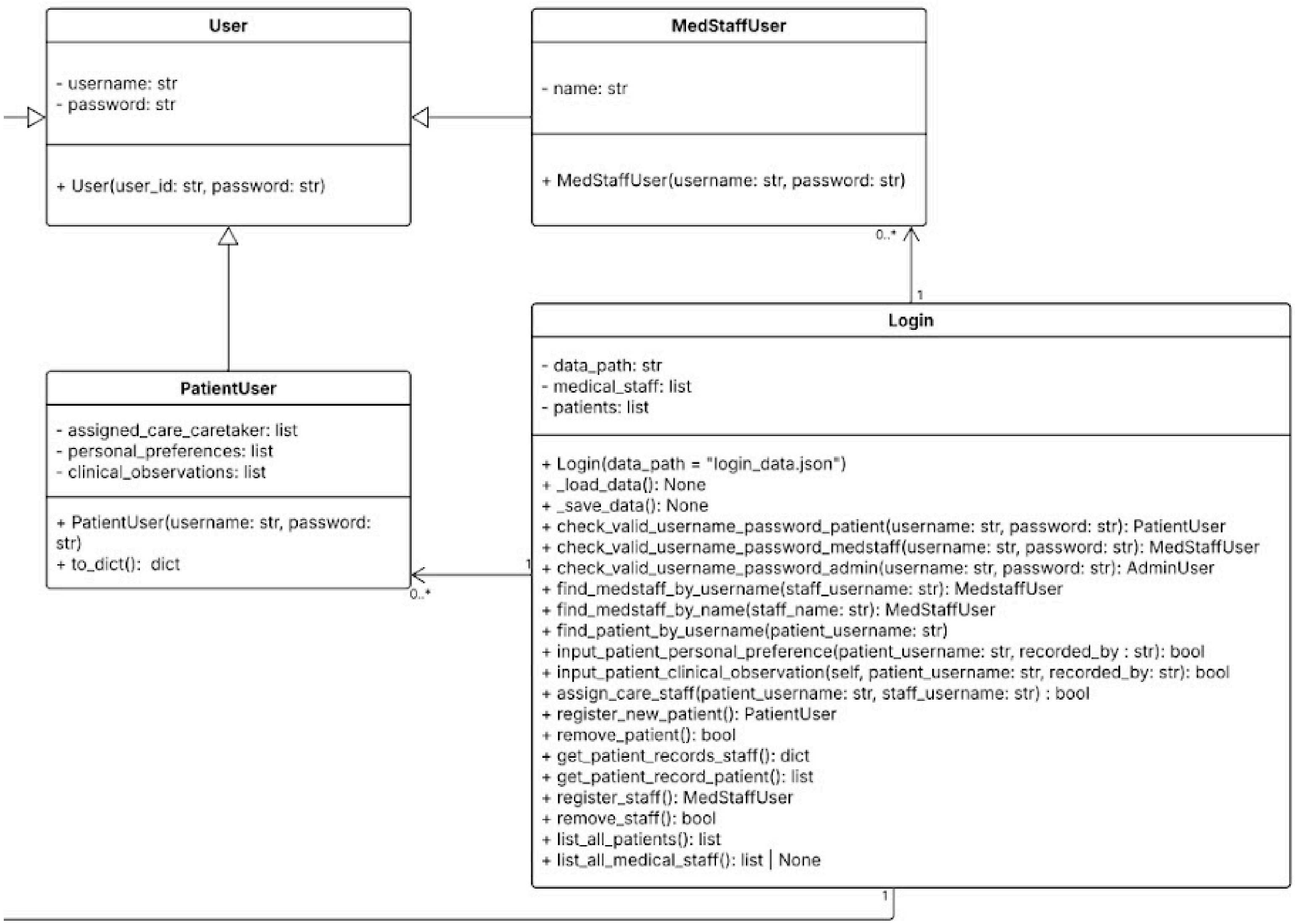
Full Class Diagram:



Partial class diagram consisting of AdminUser, PatientUser and User class:

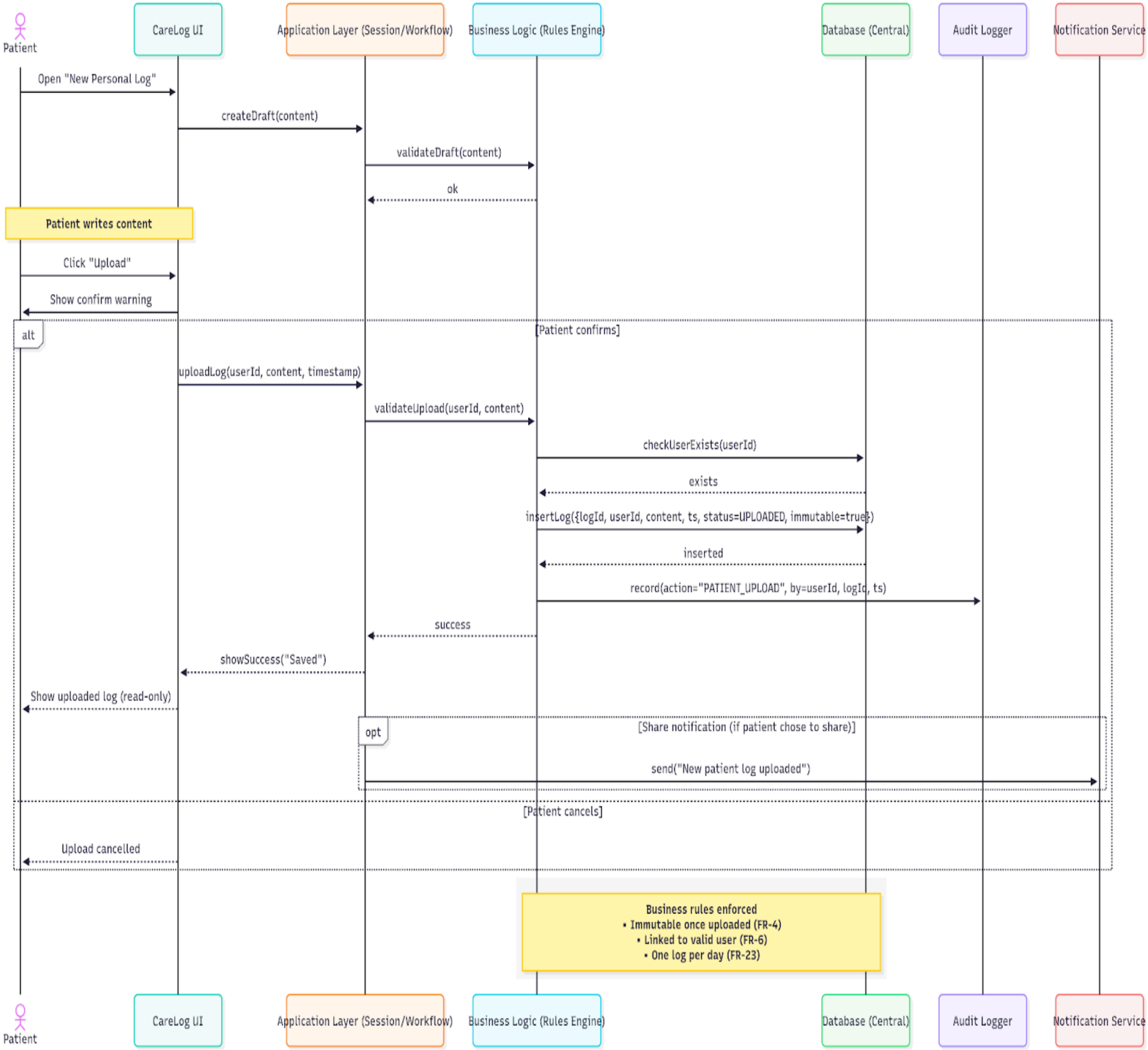


Partial class diagram consisting of User, Patient User, MedStaffUser and Login Class:

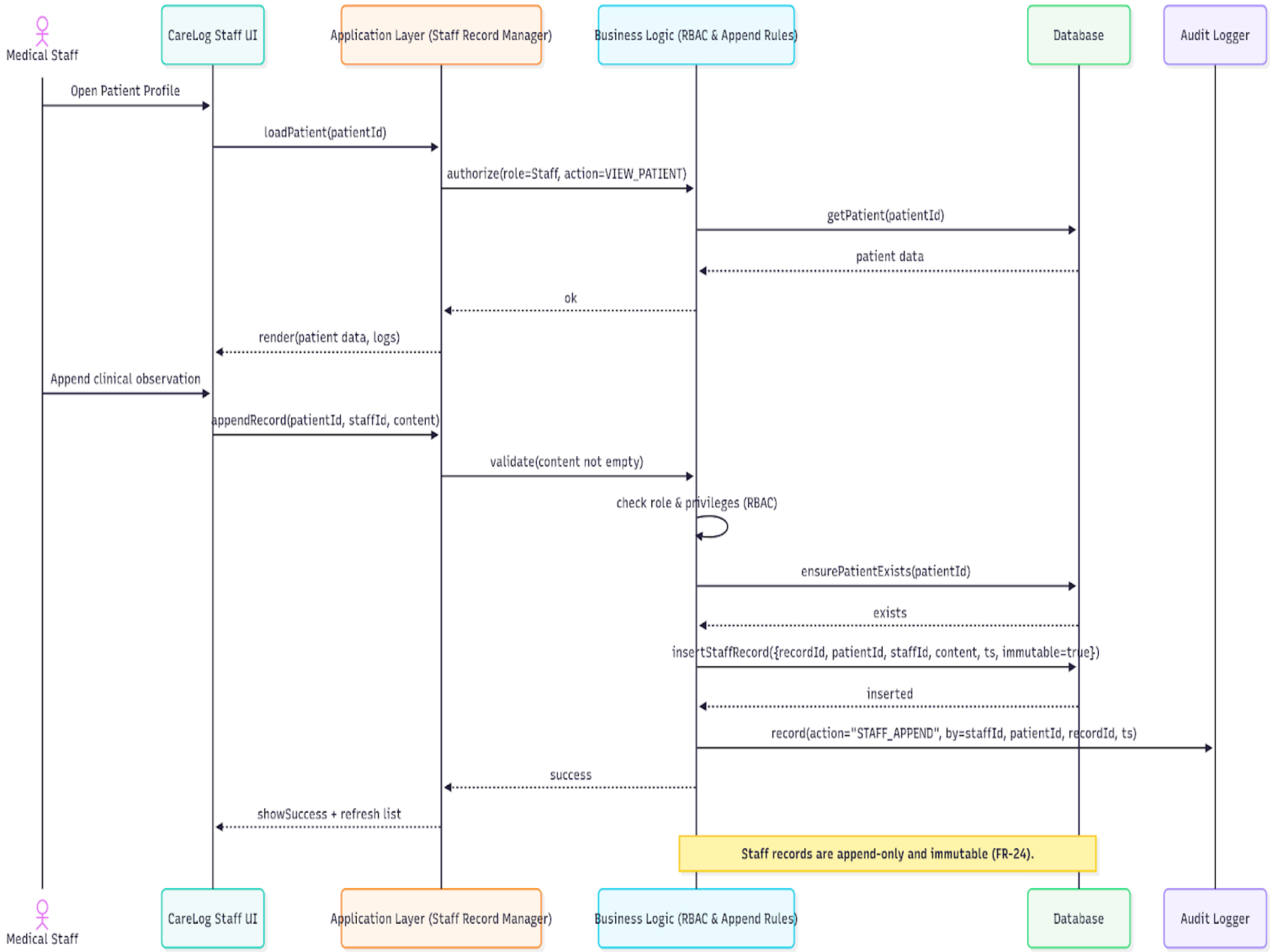


10. Sequence Diagrams

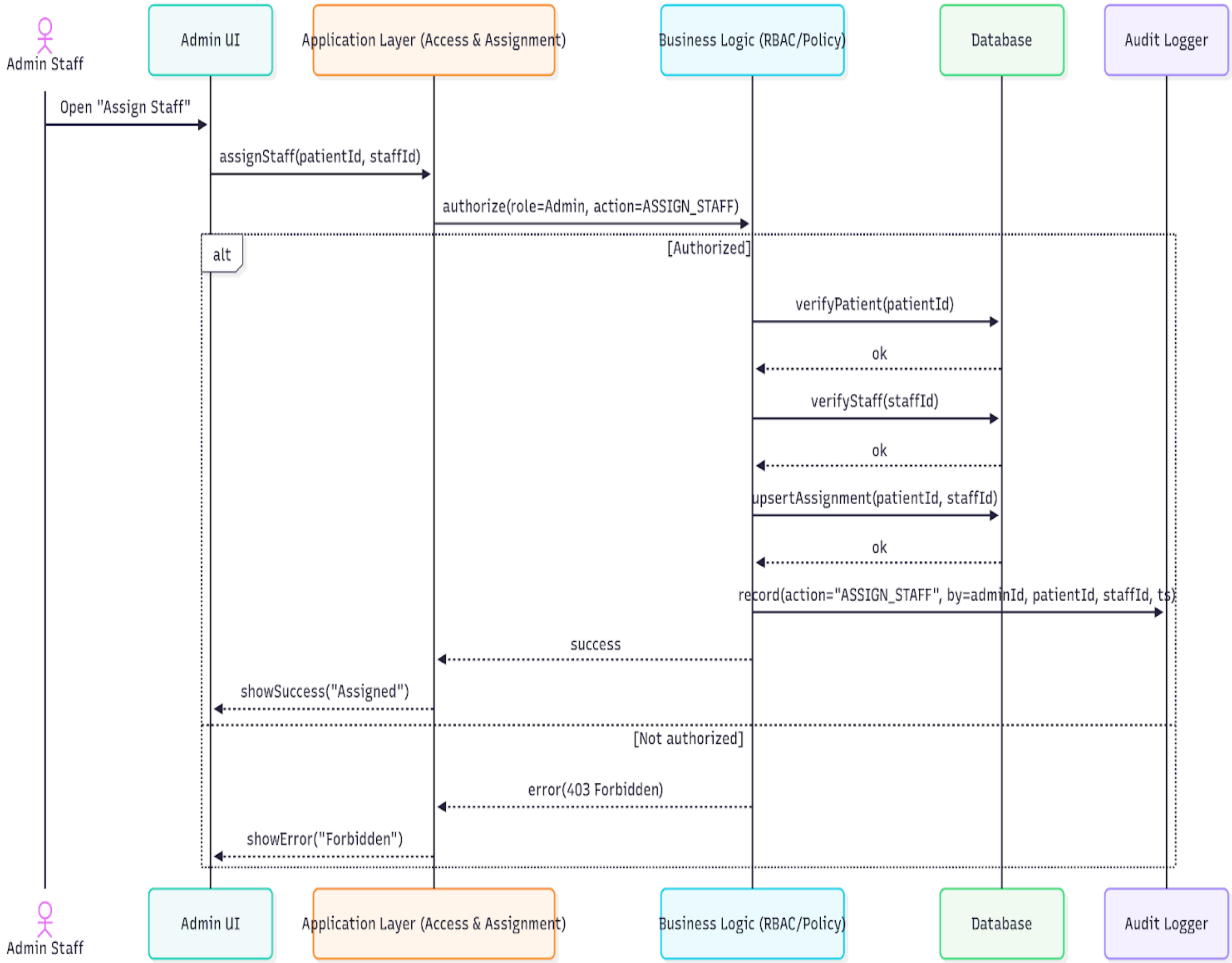
10.1 Patient Sequence Diagram:



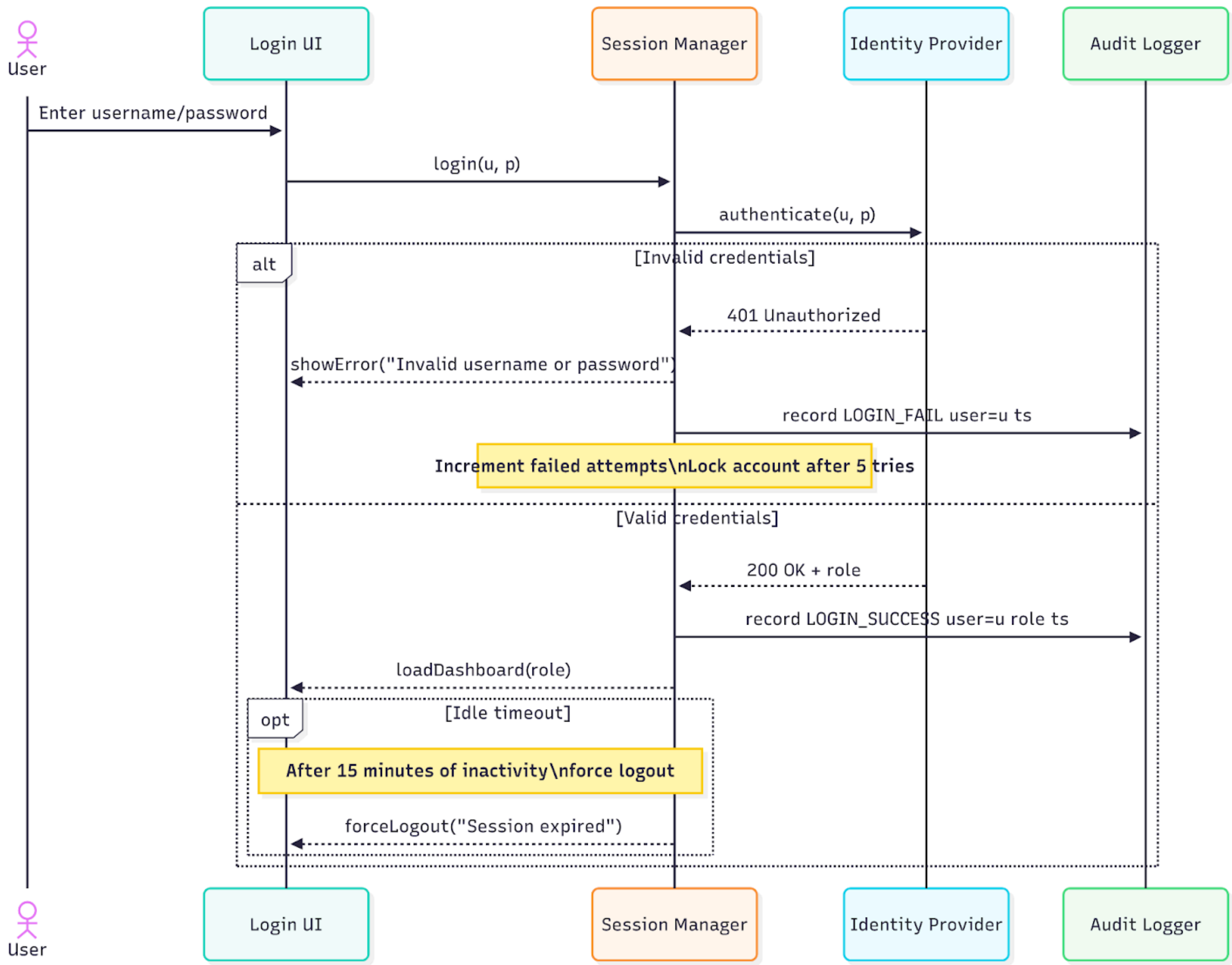
10.2 Staff Append Sequence Diagram:



10.3 Sequence Admin Diagram:



10.4 Login Sequence Diagram



11. User Evaluation

11. Effectiveness

The system demonstrates high effectiveness as users are able to achieve their goals with major bugs. This results in all users being able to perform their tasks successfully, hence 100% task success rate. No logs recorded have been lost after being saved to the database storage. The system also effectively supports all types of users to carry out their tasks successfully.

11.2 Efficiency

The system demonstrates high efficiency as users are able to perform their tasks in a short period of time e.g:

- Medical staff can create a patient clinical observation log in less than 2 minutes.
- Patients can create a personal preference log in less than 2 minutes.
- Admin can register a new patient account in less than 2 minutes.

The system provides high usability which allows users to navigate the user interface efficiently. The highly usable system is optimized to reduce time and cognitive load.

11.3 Satisfaction

Users have rated their interaction with the user interface positively (4.5/5). Some positive comments that the users have provided include "easy to use" and "system has fast response time".

11.4 Context of use

Aspect	Description
Users	Medical Staff, Admin Staff, User and Technical staff
Tasks	• Record personal preferences, • Record patient clinical observations, • Assign medical staff to patient, • Remove and register patient and medical staff accounts, • View existing patient and medical staff
Equipment	Desktop or laptop with internet access
Environment	Hospital office

11.5 User Testing

A think-aloud usability test was conducted with three respective personas: Medical staff, Admin staff and patients. All personas were used to test carry out the tasks below while stating their thoughts and challenges. All observations and outputs were collected and analyzed.

Medical Staff:

- Log patient personal preferences
- Log patient clinical observations
- View patient log record

Admin Staff

- Register new patient
- Remove existing patient
- Register new medical staff
- Remove existing medical staff
- Assign care staff to patient
- List all existing patients
- List all existing medical staff

Patient:

- Log personal preference
- View own personal preference logs

11.6 Conclusion

In conclusion, the usability evaluation found that the CareLog software system performs strongly in terms of effectiveness and satisfaction, with users being able to complete crucial tasks accurately. Minor issues affecting efficiency such as navigation clarity and lack of confirmation feedback can be addressed through interface refinements.
Overall, the system provides a user-friendly and reliable experience for its target users within the intended context of use.

12. References

Image Credits :

[1] (2025). Future-Doctor.de. https://www.future-doctor.de/wp-content/uploads/2024/08/shutterstock_2480850611.jpg

[2] (2025). Mktw.net. https://images.mktw.net/im-720565?width=1260&height=840

[3] (2025). Trios.com. https://www.trios.com/uploads/2022/06/What-Qualifications-Do-You-Need-to-be-an-Administrative-Assistant.jpg

[4] (2025). Amazonaws.com. https://informationage-production.s3.amazonaws.com/uploads/2023/08/How-to-become-a-software-developer-hero-image-1568x1018.jpg

13. Appendix

Git link: https://github.com/SnowyAurora/GP_G25

Code is located in the ver_1 branch instead of the main branch as it is only the draft version.